Predicting Building Energy Efficiency

IE 7275: Data Mining in Engineering

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Problem Setting

Buildings are among the largest contributors to urban greenhouse gas emissions, accounting for over one-third of emissions in many major cities (United Nations Environment Programme, 2025). As governments pursue climate targets, understanding what makes a building efficient and being able to identify inefficient buildings has become an essential tool for policy and urban planning.

This challenge is central to Canada's growing network of building energy benchmarking and disclosure programs. At the federal level, Natural Resources Canada (NRCan) administers ENERGY STAR Portfolio Manager, the primary tool used across the country to measure and benchmark building energy performance (Natural Resources Canada, 2025). These programs aim to create market demand for energy efficiency by making building performance transparent and supporting cities in tracking progress toward emissions-reduction goals. However, with thousands of buildings reporting each year, identifying the structural and operational factors that distinguish high-performing buildings from low-performing ones remains a significant analytical challenge.

Problem Definition

This project addresses two research questions:

1. How accurately can a building's ENERGY STAR score be predicted from its characteristics, and which features most strongly drive that score?
2. Can buildings be reliably classified as passing or failing the ENERGY STAR certification threshold (a score of 75 or above) from those same characteristics, and which features drive that classification?

These questions can help inform building energy policy by identifying which factors most strongly drive energy performance. Feature importance outputs from the classification model can be used to highlight the characteristics most associated with passing the ENERGY STAR threshold, helping building owners and policymakers take a more targeted approach to retrofits and efficiency improvements.

Data Sources & Description

The dataset used in this project will be the [2015 Building Energy Benchmarking data] published by the City of Seattle on Kaggle (City of Seattle, 2016). It contains 3,340 rows and 47 columns, with one row per building. Variables include building type, primary property type, year built, gross floor area, location, site and source energy use intensity (EUI), electricity and natural gas use, greenhouse gas emissions, and the ENERGY STAR score.

Although this dataset originates from a U.S. city, the methodology developed here is directly transferable to Canadian buildings. Both countries use the same benchmarking framework (ENERGY STAR Portfolio Manager, developed by the U.S. EPA and adapted for Canadian use by Natural Resources Canada) meaning the ENERGY STAR score, site EUI, and reported building characteristics are measured consistently across both jurisdictions.

Methodology

The dataset will first be preprocessed to ensure it is clean and suitable for analysis. This includes correcting data types, handling missing values, and normalizing variables. New variables may also be derived from existing ones where the raw data alone may not provide the information required for analysis (for example, building age).

Two supervised machine learning approaches will then be applied. The first is regression, which will be used to predict a building's Energy Star score from its energy use and physical building characteristics. The second is binary classification to identify whether a building will pass or fail Energy Star certification based on those same characteristics (omitting the Energy Star score as the label will be derived from this value). Multiple model types will be tested and compared to identify which performs best. For both approaches, the data will be split into training and testing sets to evaluate performance with unseen observations.

References

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